

HUGO HU

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EDUCATION

Stony Brook University

Stony Brook, NY

- B.S. Candidate in Computer Science | Sophomore Standing
- Honors College | Honors in Computer Science Program

May 2030

Stuyvesant High School

New York, NY

- Excellence in Mathematics Award
- Clubs: Stuyvesant Math Team, Mu Alpha Theta Math Honor Society, Spectator / Indicator Photos
- Relevant Coursework: AP Physics C (Mech & E/M), AP Calculus BC, Linear Algebra, Cybersecurity

GPA: 97.64/100

EXPERIENCE

Memorial Sloan Kettering Cancer Center

New York, NY

Student Intern, Medical Physics

May 2025 - November 2025

- Trained and improved a CNN-based deformable image registration model for aligning cone-beam CT scans to planning CTs for lung cancer treatment monitoring
- Experimented with a multi-head attention architecture for DIR, achieving performance comparable to CNN-based and VoxelMorph baselines
- Wrote MATLAB and Python preprocessing and testing code used for preparing training datasets
- Ran training and inference on HPC clusters and configured CUDA environments for training compatibility

Summer Intern

July 2024 - August 2024

- Wrote C# and Python preprocessing code for head and neck cancer CT segmentation training used in radiotherapy treatment tracking
- Used HPC clusters and IBM LSF (bsub) for job scheduling

Hack Club

Shelburne, VT and Remote

Logistics and Hardware Engineering Intern

July 2021 - March 2024

- Built a mailing address database implementing USPS Intelligent Mail Barcode (IMb), enabling no-cost letter tracking and discounted commercial postage
- Reduced domestic shipping costs by up to 60% and international shipping costs by up to 80%
- Mentored dozens of high school students through first-time PCB design reviews
- Designed open-source hardware including a 2D CNC motor controller, USB-C Power Delivery supply, and handheld gaming console
- Published open-source reference designs and documentation for NFC tags and USB hubs
- Managed the manufacture and global mailing of 1000+ hardware kits
- Helped organize in-person hackathons in New York City, Vermont, and San Francisco

PROJECTS

Personal Site Platform | NextJS, Supabase, Cloudinary CDN | [GitHub](#)

2026

- **Personal Site and Photography Portfolio:** Built a Next.js personal website with a dynamic photography portfolio, NextAuth-secured admin dashboard, and shared SQL APIs
- **Component Inventory Platform:** Built a secure inventory manager with SQL-backed search, barcode scanning, PDF label generation, and component management

SAMD21 Corginator | KiCAD, CircuitPython | [GitHub](#)

2024

- Designed a corgi-shaped ATSAM21 development board with e-marked USB-C compatibility and support for Arduino IDE and CircuitPython

SKILLS & AWARDS

Languages: C, C++, Python, MATLAB, C#, Java

Frameworks: Next.js, HTML/CSS, Supabase

Tools: Git, Linux, KiCAD, LaTeX

Laboratory: Soldering, reflow, and rework equipment

Awards: AIME Qualifier (3×), CEMC Euclid/Fermat School Winner (2025), Gold President's Volunteer Service Award